

Before you send this

EIBIS Firm-Level Microdata — Research Proposal — Draft prepared by TWIN Analytics, September 2026

Where to submit: email the completed proposal (with CV attached) to **EIBIS_data_access@eib.org**. This is an email-based process, not an online form — EIB does not publish a web submission portal for this. Background on the access process: <https://www.eib.org/en/publications-research/economics/surveys-data/eibis/about/index>

What still needs your input

The document on the following pages is drafted from the analysis already completed in the TWIN_Analytics repository and can be used close to as-is. Six fields are marked in orange and need your input before this is ready to send:

- Your name(s), institutional affiliation, role, and contact email (§1).
- Collaborators or institutional partners, if any (§1) — EIBIS access is granted “for collaborative projects aiming at a publication,” so this matters.
- A CV, attached separately (§1) — stated as a requirement by EIBIS.
- The specific panel years you're requesting and whether your institution has separate ORBIS access (§5) — the ORBIS question materially strengthens the proposal since it's the remaining link needed for a complete firm-level chain.
- Your intended publication venue or output format (§6) — this appears to be a real gating criterion for the committee, not boilerplate.
- A timeline (§7).

Two things worth doing before you send it

- **Attach the existing technical report** as evidence of research capacity: *TWIN_Analytics_First_Findings.pdf* (2-page summary) and/or a link to https://github.com/FIMPlank/TWIN_Analytics. The aggregate-level work is real, independently reviewed across five rounds, and directly motivates the access being requested — don't undersell it.
- **If your institution does not have ORBIS access**, RQ2/H2 (the firm-level verified-emissions link) will need EIBIS's own guidance on whether their microdata already carries a usable firm identifier that maps to EU ETS installations, or whether a different linking strategy is needed. Worth asking directly in the email rather than assuming.

EIBIS Firm-Level Microdata

Research Proposal

1. Applicant(s)

[FILL IN] Name(s)

[FILL IN] Institutional affiliation

[FILL IN] Position / role

[FILL IN] Contact email

CV(s): attached separately, per EIBIS's stated requirement.

[FILL IN] Collaborators, if any — EIBIS microdata access is granted “for collaborative projects aiming at a publication,” so note co-authors or institutional partners here.

2. Project title

Twin Transformation: Does Digitalization Predict Verified Emissions Reduction, or Only Disclosed Climate Ambition? A Firm-Level Test

3. Background and motivation

The EU's “twin transformation” agenda treats the digital and green transitions as mutually reinforcing. This is a plausible but largely untested empirical claim at firm level: does a firm's digital-technology adoption predict a reduction in its *actual, independently verified* emissions, or only in its *disclosed* climate targets and investment intentions — measures that are self-reported and not independently audited?

We have already completed an extensive country- and sector-level test of this question using open data (EU ETS verified emissions, EIBIS *aggregate* survey results, Eurostat's Digital Intensity Index, and an independent US state/county extension using EPA facility-level emissions data), summarized in the attached technical report and at https://github.com/FIMPlank/TWIN_Analytics. That analysis — built and independently reviewed across multiple rounds of researcher/reviewer scrutiny, with small-cluster-robust inference, placebo/timing checks, and convergence-economics robustness tests — found **no significant relationship** between digitalization intensity and verified-emissions change, at the country, industrial-sector, or (in a supplementary regional test) sub-national level, across an 11-year EU panel, two independent digitalization measures, and a parallel US analysis.

This null result is informative but structurally limited: **every variable in that analysis is a country-, sector-, or region-wide average**. An aggregate-level correlation (or its absence) cannot distinguish between two very different underlying realities — (a) digitalizing firms and emissions-cutting firms are genuinely unrelated populations, or (b) they are the *same* firms, but the relationship is invisible at the aggregate level because it is diluted by averaging across firms that never digitalized and firms that were never significant emitters in the first place. Only firm-level microdata that observes *the same firms* on both a digitalization measure and a climate/environmental-investment measure can distinguish these.

EIBIS is, to our knowledge, the best-positioned dataset to close this gap: a panel since 2016, ~13,000 EU+US firms/year, with **both a digitalization module and a climate/environmental investment module administered to the same respondents in the same survey wave** — precisely the co-observation that no open dataset we have found provides.

4. Research objectives and hypotheses

RQ1 (primary): At the firm level, does digital-technology adoption intensity predict a firm's *disclosed climate investment or GHG-target-setting behavior* (EIBIS's own climate module — the “disclosure” side)?

RQ2 (primary, the gap this proposal is designed to close): Where EIBIS firm identifiers can be linked to EU ETS installation-level *verified* emissions data (via the JRC-EU-ETS-FIRMS crosswalk to ORBIS firm IDs — already inspected in our open-data work, but not linkable to EIBIS without this access), does digitalization intensity predict *actual verified emissions change*, not just disclosed intent?

H1: Firm-level digitalization intensity is positively associated with disclosed climate investment/target-setting (a relatively low bar, testing survey-response consistency).

H2 (the substantive test): Firm-level digitalization intensity is associated with *verified* emissions reduction, net of firm size, sector, and country fixed effects.

H3 (the disclosure-reality gap, the project's central question): The disclosure-side association (H1) is stronger/more robust than the verified-outcome association (H2) — i.e., digitalization predicts what firms say about their climate ambitions more reliably than what independently audited data shows they *do*.

Our aggregate-level prior (a robust null on the verified-outcome side, replicated across country, sector, regional, and cross-national US tests, not yet tested against a disclosure-side benchmark at the same level of aggregation) is directly consistent with H3 but cannot distinguish it from simple non-existence of any relationship. Firm-level data is required to test H3 as stated.

5. Data and methodology

Primary data requested: EIBIS firm-level microdata (digitalization module + climate/environmental investment module), EU+US coverage, panel years:

[FILL IN] specify range, e.g. 2018-2025 if available, or match to survey waves overlapping EU ETS phase 4

Linked data (already open-access, already integrated into our pipeline):

- EU ETS verified emissions (installation-level, EUTL) — already downloaded and validated in our repository.
- JRC-EU-ETS-FIRMS (EU ETS account holder ↔ ORBIS/Bureau van Dijk firm-ID crosswalk) — already downloaded; this is the linking key that would let EIBIS firm IDs be matched to verified ETS emissions, if EIBIS provides (or can be matched to) ORBIS/BvD firm identifiers.

[FILL IN] Does your institution have separate ORBIS financial/firmographic access? This materially strengthens the proposal, since it is the remaining link needed for a complete firm-level chain.

Methodology: Firm-level panel regression (fixed effects: firm, sector, country, year), with the same standard of statistical rigor already demonstrated in our open-data work: cluster-robust standard errors appropriate to the panel's cluster structure (with small-sample corrections, e.g. wild cluster bootstrap, where the number of clusters is limited), a placebo/timing check (does future digitalization “predict” past emissions? — a check for reverse causality), and controls for firm size, sector composition, and macroeconomic conditions (GDP growth, energy prices) drawn from the same free public sources already integrated into our pipeline (Eurostat).

Reproducibility and transparency: All non-restricted code, data, and documentation for this project are maintained in a public repository (https://github.com/FIMPlank/TWIN_Analytics). We would maintain the same standard for any firm-level extension, publishing all derived/aggregated results and code subject to EIBIS's microdata confidentiality and disclosure-control requirements, and would follow EIBIS's stated access modality (on-site/remote secure environment, as required) for any non-aggregable output.

6. Expected outputs / publication plan

[FILL IN] EIBIS microdata access is stated to be granted “for collaborative projects aiming at a publication.” Specify your intended venue/format: a working paper, a submission to a specific journal or conference, a policy report, etc. If this is an academic project, name the target outlet or research program it sits within.

7. Timeline

[FILL IN] e.g. data access requested by [DATE], analysis over [N months], draft results by [DATE].